

Wei TANG

Curriculum Vitae



PERSONAL DATA

PLACE AND DATE OF BIRTH	Yancheng, China 08 December 1993
NATIONALITY	Chinese
WORK ADDRESS	Department of Physics and Astronomy, Ghent University, Krijgslaan 281, 9000 Ghent, Belgium
EMAIL	wei.tang.phys@gmail.com
ONLINE ACADEMIC PROFILE	Google Scholar ORCID

ACADEMIC POSITIONS

PRESENT	FWO Senior Postdoc, Ghent University, Ghent, Belgium
OCT. 2024	Advisor: Prof. Jutho HAEGEMAN, Prof. Frank VERSTRAETE
SEPT. 2024	Postdoc, Ghent University, Ghent, Belgium
OCT. 2021	Advisor: Prof. Jutho HAEGEMAN

EDUCATION

JULY 2021	Ph.D. in Physics , Peking University, Beijing, China
SEPT. 2016	Major: Condensed Matter Physics Advisor: Prof. Xin-Cheng XIE and Prof. Lei WANG
AUG. 2019	Visiting Ph.D. student, Technical University of Dresden, Dresden, Germany
FEB. 2019	Advisor: Prof. Hong-Hao TU
AUG. 2018	Visiting Ph.D. student, Technical University of Dresden, Dresden, Germany
JUNE 2018	Advisor: Prof. Hong-Hao TU
JUNE 2016	B.S. in Physics , Nanjing University, Nanjing, China
SEPT. 2012	Major: Physics

ACADEMIC SERVICES

Referee for Physical Review Letters, Physical Review B, Physical Review Research, Communications Physics, Journal of Physics A: Mathematical and Theoretical.

SCHOLARSHIPS AND AWARDS

JUN 2024	FWO Senior Postdoc Fellowship, the Research Foundation – Flanders
FALL 2020	Specialized Scholarship for Graduate students, Peking University
FALL 2019	Specialized Scholarship for Graduate students, Peking University
FALL 2017	Specialized Scholarship for Graduate students, Peking University
FALL 2013-2015	Renmin Scholarship, Nanjing University

TEACHING EXPERIENCE

FALL 2021 Teaching Assistant, Python for Scientists, Ghent University
FALL 2017 Teaching Assistant, Quantum Mechanics, Peking University

PUBLICATIONS

- [1] **Wei Tang**, Benoît Tuybens, Jutho Haegeman, *Numerical study of boson mixtures with multi-component continuous matrix product states*, [arXiv:2512.24998](#) (2025).
- [2] Xing-Yu Zhang, Qi Yang, Philippe Corboz, Jutho Haegeman, **Wei Tang**, *Accelerating two-dimensional tensor network optimization by preconditioning*, [arXiv:2511.09546](#) (2025).
- [3] **Wei Tang**, Laurens Vanderstraeten, Jutho Haegeman, *Gauging the variational optimization of projected entangled-pair states*, [arXiv:2508.10822](#) (2025).
- [4] Kevin Vervoort, **Wei Tang**, Nick Bultinck, *Extracting average properties of disordered spin chains with translationally invariant tensor networks*, [arXiv:2504.21089](#) (2025).
- [5] Bi-Yang Tan, Yueshui Zhang, Hua-Chen Zhang, **Wei Tang**, Lei Wang, Hong-Hao Tu, Ying-Hai Wu, *Extracting the Luttinger parameter from a single wave function*, *Phys. Rev. Lett.* **134**, 076501 (2025).
- [6] **Wei Tang**, Frank Verstraete, Jutho Haegeman, *Matrix product state fixed points of non-Hermitian transfer matrices*, *Phys. Rev. B* **111**, 035107 (2025).
- [7] **Wei Tang**, Jutho Haegeman, *Kac-Moody symmetries in one-dimensional bosonic systems*, *Phys. Rev. B* **108**, 035153 (2023).
- [8] Yueshui Zhang, Anton Hulsch, Hua-Chen Zhang, **Wei Tang**, Lei Wang, Hong-Hao Tu, *Universal Scaling of Klein Bottle Entropy near Conformal Critical Points*, *Phys. Rev. Lett.* **130**, 151602 (2023).
- [9] **Wei Tang**, X. C. Xie, Lei Wang, Hong-Hao Tu, *Tensor network simulation of the $(1+1)$ -dimensional $O(3)$ nonlinear σ -model with $\theta = \pi$ term*, *Phys. Rev. D* **104**, 114513 (2021).
- [10] **Wei Tang**, Hong-Hao Tu, Lei Wang, *Continuous matrix product operator approach to finite temperature quantum states*, *Phys. Rev. Lett.* **125**, 170604 (2020).
- [11] **Wei Tang**, X. C. Xie, Lei Wang, Hong-Hao Tu, *Quantized thermal Hall conductance from edge current calculations in lattice models*, *Phys. Rev. B* **100**, 155112 (2019).
- [12] **Wei Tang**, X. C. Xie, Lei Wang, Hong-Hao Tu, *Klein bottle entropy of compactified boson conformal field theory*, *Phys. Rev. B* **99**, 115105 (2019).
- [13] **Wei Tang**, Lei Chen, Wei Li, X. C. Xie, Hong-Hao Tu, Lei Wang, *Universal boundary entropies in conformal field theory: a quantum Monte Carlo study*, *Phys. Rev. B* **96**, 115136 (2017), Editors' suggestion.

SEMINAR TALKS

Invited Seminars

- DEC. 2024 *Matrix product state fixed points of non-Hermitian transfer matrices*, Fudan University, Shanghai, China
- FEB. 2024 *Matrix product state fixed points of non-Hermitian transfer matrices*, Instituto de física fundamental, Consejo Superior de Investigaciones Científicas (CSIC), Madrid, Spain
- JAN. 2024 *Matrix product state fixed points of non-Hermitian transfer matrices*, Ludwig Maximilian University of Munich, Munich, Germany
- DEC 2023 *Matrix product state fixed points of non-Hermitian transfer matrices*, Institute of Physics, Chinese Academy of Sciences, Beijing, China

- JUNE 2020 *Continuous matrix product operator approach to finite temperature quantum states,*
(Online) Ghent University, Ghent, Belgium
- AUG. 2018 *Universal entropy of conformal critical theories on a Klein bottle,*
University of Würzburg, Würzburg, Germany
- JUL. 2018 *Universal entropy of conformal critical theories on a Klein bottle,*
Ludwig Maximilian University of Munich, Munich, Germany
- JUNE 2018 *Universal entropy of conformal critical theories on a Klein bottle,*
Technical University of Dresden, Dresden, Germany
- MAY 2018 *Universal entropy of conformal critical theories on a Klein bottle,*
Chongqing University, Chongqing, China

Regular Seminars

- FEB. 2024 *Matrix product state fixed points of non-Hermitian transfer matrices,*
Ghent University, Ghent, Belgium

PARTICIPATION IN CONFERENCES/WORKSHOPS

Invited Talks

- DEC. 2024 *Matrix product state fixed points of non-Hermitian transfer matrices*
Tensor Network States: Algorithms and Applications (TNSAA)
Hong Kong, China

Contributed Talks

- JAN. 2026 *Gauging the variational optimization of projected entangled-pair states*
Fixed-Point Optimization in Tensor Networks for Quantum Matter,
Zürich, Switzerland
- OCT. 2025 *Gauging the variational optimization of projected entangled-pair states*
A Panorama of Tensor Networks: from Condensed Matter to Quantum Field
Theory, Holography and Beyond,
Krakow, Poland
- APR. 2024 *Matrix product state fixed points of non-Hermitian transfer matrices*
TUM/IAS Workshop: Recent progress on tensor network methods,
Munich, Germany
- AUG. 2023 *Kac-Moody symmetries in one-dimensional bosonic systems*
Entanglement in Strongly Correlated Systems,
Benasque, Spain
- JAN. 2022 *Continuous matrix product operator approach to finite temperature quantum states and its application to the nonlinear sigma model with $\theta = \pi$ term*
Tensor Network States: Algorithms and Applications (TNSAA),
Online

Poster contribution

- APR. 2024 *Matrix product state fixed points of non-Hermitian transfer matrices*
Topological order: Anyons and Fractons,
Les Houches, France
- FEB. 2021 *Continuous matrix product operator approach to finite temperature quantum states*
Entanglement in Strongly Correlated Systems,
(Online) Benasque, Spain
- DEC. 2020 *Continuous matrix product operator approach to finite temperature quantum states*
Winter School on Strongly Correlated Quantum Matter,
(Online) Dresden, Germany
- AUG. 2019 *Quantized thermal Hall conductance from edge current calculations in lattice models*

- Retreat Meeting 2019 of SFB 1143,
SFB 1143, Nimbschen, Germany
- JUNE 2019 *Quantized thermal Hall conductance from edge current calculations in lattice models*
One-Dimensional Systems for Quantum Technology,
Physikzentrum Bad Honnef, Bad Honnef, Germany
- JUNE 2019 *Quantized thermal Hall conductance from edge current calculations in lattice models*
Topological Quantum Field Theory and Quantum Hall Physics,
Physikzentrum Bad Honnef, Bad Honnef, Germany
- APR. 2018 *Universal entropy of conformal critical theories on a Klein bottle*
The 8th Workshop on Quantum Many-Body Computation,
Zhejiang University, Hangzhou, China
- APR. 2018 *Universal entropy of conformal critical theories on a Klein bottle*
The 2nd International Workshop on Critical Behavior in Lattice Models,
Anqing Normal University, Anqing, China

Attendance

- FEB. 2025 Entanglement in Strongly Correlated Systems,
Benasque, Spain
- OCT. 2022 Mathematical structure of Tensor Networks,
Erwin Schrödinger Institute, Vienna, Austria
- JULY 2021 International workshop on tensor networks in many body and lattice field,
Tsung-Dao Lee Institute, Shanghai, China
- AUG. 2020 Higher Structures and Field Theory,
(Online) Erwin Schrödinger Institute, Vienna, Austria
- AUG. 2018 Retreat Meeting 2018 of SFB 1143,
SFB 1143, Meissen, Germany
- JUNE 2018 Machine Learning for Quantum Many-body Physics,
Max Planck Institute for the Physics of Complex Systems, Dresden, Germany
- JAN. 2018 Tensor Network States: Algorithms and Applications,
Institute of Physics, Chinese Academy of Sciences, Beijing, China
- JUNE 2017 Machine Learning and Many-Body Physics,
Kavli Institute for Theoretical Sciences, University of Chinese Academy of Sciences, Beijing, China
- JUNE 2017 Strongly Interacting Conformal Field Theory in Condensed Matter Physics,
Tsinghua University, Beijing, China
- AUG. 2016 Computational Approaches for Quantum Many Body Systems,
University of Chinese Academy of Sciences, Beijing, China

REFERENCE

- **Prof. Jutho Haegeman**
Department of Physics and Astronomy, Ghent University
Ghent 9000, Belgium
E-mail: Jutho.Haegeman@UGent.be
- **Prof. Frank Verstraete**
Department of Applied Mathematics and Theoretical Physics, University of Cambridge
Cambridge CB2 1TN, United Kingdom &
Department of Physics and Astronomy, Ghent University
Ghent 9000, Belgium
E-mail: frank.verstraete@ugent.be
- **Prof. Lei Wang**
Beijing National Lab for Condensed Matter Physics and Institute of Physics
Chinese Academy of Sciences
Beijing 100190, China

E-mail: wanglei@iphy.ac.cn

- **Prof. Hong-Hao Tu**

Faculty of Physics, Ludwig-Maximilians-University

Munich 80333, Germany

Tel: +49 (0)89 / 2180 - 4539

E-mail: h.tu@lmu.de

- **Prof. Xin-Cheng Xie**

International Center for Quantum Materials, School of Physics, Peking University

Beijing 100871, China

E-mail: xcxie@pku.edu.cn